

Heart Disease Prediction

A PROJECT REPORT

Submitted by

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CERTIFICATE

This is to certify that the INTERNSHIP project entitled “**Heart Disease Prediction**” Have been carried out by **Akanksha Vaghela (230343131023)** under my guidance in **COMPUTER SCIENCE AND ENGINEERING (8-SEM)** of Gujarat Technological University, Ahmedabad during the academic year 2026.

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ACKNOWLEDGEMENT

I, the developer of the project “Heart Disease Prediction”, with immense pleasure and commitment would like to present the project assignment. The development of this project has given me wide opportunity to think, implement and interact with various aspects of management skills as well as the new emerging technologies.

Every work that one completes successfully stands on the constant encouragement, good will and support of the people around. I hereby avail this opportunity to express my gratitude to number of people who extended their valuable time, full support and cooperation in developing the project.

I express deep sense of gratitude towards our Head of the CSE Department, Prof. Parth Goel and project guide Prof. Nilesh Dubey for the support during the whole session of study and development. It is because of them, that I was prompted to do hard work, adopting new technologies.

I would also like to thank all the mentor for their guidelines throughout the development phase of the project. They encouraged me to look forward to learn and implement new and emerging technologies.

They also guided me to go for some user friendly and extremely useful real-life application.

They altogether provided me favorable environment, and without them it would not have been possible to achieve my goal.

ABSTRACT

Such information like Heart Disease, if predicted well in advance, can provide important insights to doctors who can then adapt their diagnosis and treatment per patient basis. Disease prediction using machine learning is extremely much useful in everyone's day to day life and it's mainly more important for the healthcare sector, because they're the one that daily uses these systems to predict the diseases of the patients supported their general information and there symptoms that they're been through.

This project Heart Disease prediction using machine learning is extremely much useful in everyone's day to day life and it's mainly more important for the healthcare sector, because they're the one that daily uses these systems to predict the diseases of the patients supported their general information and there symptoms that they're been through. Now a day's health industry plays important role in curing the diseases of the patients so this is often also some quite help for the health industry to inform the user and also it's useful for the user just in case he/she doesn't want to travel to the hospital or the other clinics, so just by entering the symptoms and every one other useful information within the form user can get to know the disease he/she is affected by and therefore the health industry also can get enjoy this technique by just asking the symptoms from the user and entering within the system and in only few seconds they will tell the precise and up to some extent the accurate diseases.

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CHAPTER 1: PROJECT DEFINITION

**To create Python application which helps in
detecting Heart Disease**

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CHAPTER 2:

PROJECT DESCRIPTION

2.1 PROJECT OVERVIEW:

Such information like Heart Disease, if predicted well in advance, can provide important insights to doctors who can then adapt their diagnosis and treatment per patient basis. Now a day's health industry plays important role in curing the diseases of the patients so this is often also some quite help for the health industry to inform the user and also it's useful for the user just in case he/she doesn't want to travel to the hospital or the other clinics, so just by entering the symptoms and every one other useful information within the form user can get to know the disease he/she is affected by and therefore the health industry also can get enjoy this technique by just asking the symptoms from the user and entering within the system and in only few seconds they will tell the precise and up to some extent the accurate diseases.

If health industry adopts this project then the work of the doctors are often reduced and that they can easily predict the disease of the patient. The Disease prediction is to supply prediction for the varied and usually occurring diseases that when unchecked and sometimes ignored can turns into fatal disease and cause lot of problem to the patient. We can update this project in future by adding more attributes to the dataset and more interactive to the users and can also be done as a mobile application. we will modify the system by connecting it to the hospital's database.

2.2 INTRODUCTION TO DOMAINS:

MACHINE LEARNING

In this post we'll provide a general introduction to machine learning, which tries to highlight the underlying technical challenges and where we have solutions. Machine learning is the principle technology underpinning the recent advances in artificial intelligence.

Machine learning is perhaps the principal technology behind two emerging domains: data science and artificial intelligence. The rise of machine learning is coming about through the availability of data and computation, but machine learning methodologies are fundamentally dependent on models.

Data + Model → Prediction

The emergence of machine learning is closely tied to the emergence of widely available data.

COMPUTER VISION

Computer vision is the field of computer science that focuses on replicating parts of the complexity of the human vision system and enabling computers to identify and process objects in images and videos in the same way that humans do. Until recently, computer vision only worked in limited capacity.

Thanks to advances in artificial intelligence and innovations in deep learning and neural networks, the field has been able to take great leaps in recent years and has been able to surpass humans in some tasks related to detecting and labelling objects.

CHAPTER 3:
SOFTWARE AND HARDWARE
ENVIRONMENT

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3.1 HARDWARE AND SOFTWARE THAT WERE USED IN THIS PROJECT:-

SOFTWARE USED:

- VSCode
- Chrome Browser
- PyCharm

SYSTEM CONFIGURATION ON WHICH THE PROJECT WAS DEVELOPED:

- Laptop/computer with Core i5 8TH gen. Processor
- 2BG Hard Disk Storage
- 8 GB RAM

OTHER REQUIREMENTS:

- Python
- KNN

CHAPTER 4:
MAJOR FUNCTIONALITY

4.1 Main Window

The screenshot shows a desktop application window titled "Heart Disease Prediction". The window is divided into two main sections. The left section, titled "Fill your Details", contains a form with the following fields and options:

- Age: [0]
- Sex: ☐ Male ☒ Female
- Chest pain type (0-3): [0]
- Resting blood pressure: [0]
- Cholestoral (mg/dl): [0]
- Blood sugar > 120(mg/dl): ☐ True ☒ False
- Resting Electrocardiographic: ☒ 0 ☐ 1 ☐ 2
- Max Heart rate: [0]
- Pain during exercise: ☐ Yes ☒ No
- ECG entry rate: [0.0]
- ECG peak: ☒ upsloping ☐ flat ☐ downsloping
- vessels colored by flourosop: ☒ 0 ☐ 1 ☐ 2 ☐ 3
- Thalassemia: ☒ 0 ☐ 1 ☐ 2 ☐ 3

At the bottom of the form is a "Predict" button. The right section of the window features a large, 3D-rendered red heart with a silver stethoscope draped around it.

Figure 1 Main Window

On visiting our desktop application, this is the Main Page that will be seen by the user which consists of a general form which will ask user about the basic input.

The input consist of:

- Age
- Sex

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- Chest Pain Type (0-3)
- Resting Blood Pressure
- Cholestoral (mg/dl)
- Blood Sugar
- Resting Electrocardiographic
- Max Heart rate
- Pain during exercise (Binary)
- ECG entry rate
- ECG peak

- vessels colored by flourosop

4.2 Healthy Page



Figure 2 Healthy Page

After the successful input of the user, he/she will be directed to the **Healthy Page** if the person is predicted healthy and will be able to see this kind of window.

This says “**Person has a low chance of having a Heart Disease**”

4.3 Unhealthy Page



Figure 3 Unhealthy Page

After the successful input of the user, he/she will be directed to the **Unhealthy Page** if the person is predicted unhealthy and will be able to see this kind of window.

This says “**Person has a high chance of having a Heart Disease**”

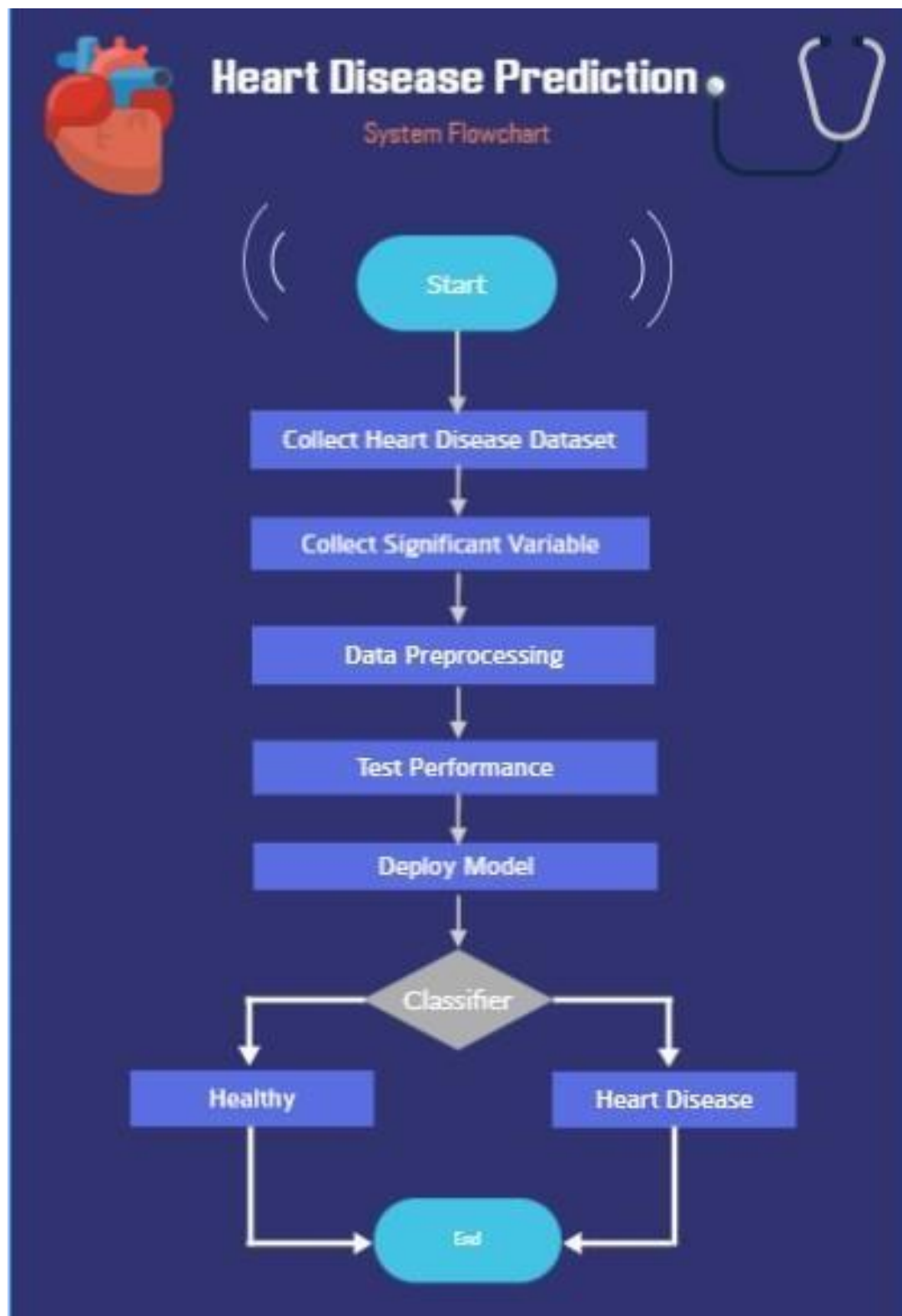


Figure 4 System Flowchart

CHAPTER 6:
IMPLEMENTATION SNAPSHOTS OF
PROJECT

Heart Disease Predictor

Fill your Details

Age

0

Sex

☐ Male

☒ Female

Chest pain type (0-3)

0

Resting blood pressure

0

Cholestoral (mg/dl)

0

Blood sugar > 120(mg/dl)

☐ True

☒ False

Resting Electrocardiographic

☒ 0

☐ 1

☐ 2

Max Heart rate

0

Pain during exercise

☐ Yes

☒ No

ECG entry rate

0.0

ECG peak

☒ upsloping

☐ flat

☐ downsloping

vessels colored by flourosop

☒ 0

☐ 1

☐ 2

☐ 3

Thalassemia

☒ 0

☐ 1

☐ 2

☐ 3

Predict

Heart Disease Prediction



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Figure

6 Predicted Healthy

After the successful input of the user, he/she will be directed to the **Healthy Page** if the person is predicted healthy and will be able to see this kind of window.

This says “**Person has a low chance of having a Heart Disease**”



Figure 7 Predicted Unhealthy

After the successful input of the user, he/she will be directed to the **Healthy Page** if the person is predicted healthy and will be able to see this kind of window.

This says “**Person has a high chance of having a Heart Disease**”

CHAPTER 7: **PROJECT LIMITATIONS**

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Accuracy

Accuracy is not up to the mark as it should be comparing to other similar project available in the market.

Time Consuming

When the dataset size is bigger then it could take a little bit more time.

Particular Disease

These version only tells that a person has heart disease or not but it should provide with any kind of particular disease detected.

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CHAPTER 8: **PROJECT OUTCOMES**

We successfully implemented the Heart Disease Prediction to allow the user to identify any kind of Heart disease is there or not by asking some basic input which can be collected by any laboratory

Machine learning disease prediction is a system that predicts diseases based on information provided by users. It predicts the disease of the patient or the user based on the information or the symptoms enter into the web system and gives results based on that information. If the patient isn't much serious and therefore the user just wants to understand the sort of disease, he/she has been through.

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CHAPTER 9: **FUTURE ENHANCEMENTS**

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More Accuracy

We will try to improve our accuracy score, comparing to our current accuracy score to predict the disease.

Time Complexity

Time complexity of searching can be reduced by using clustering.

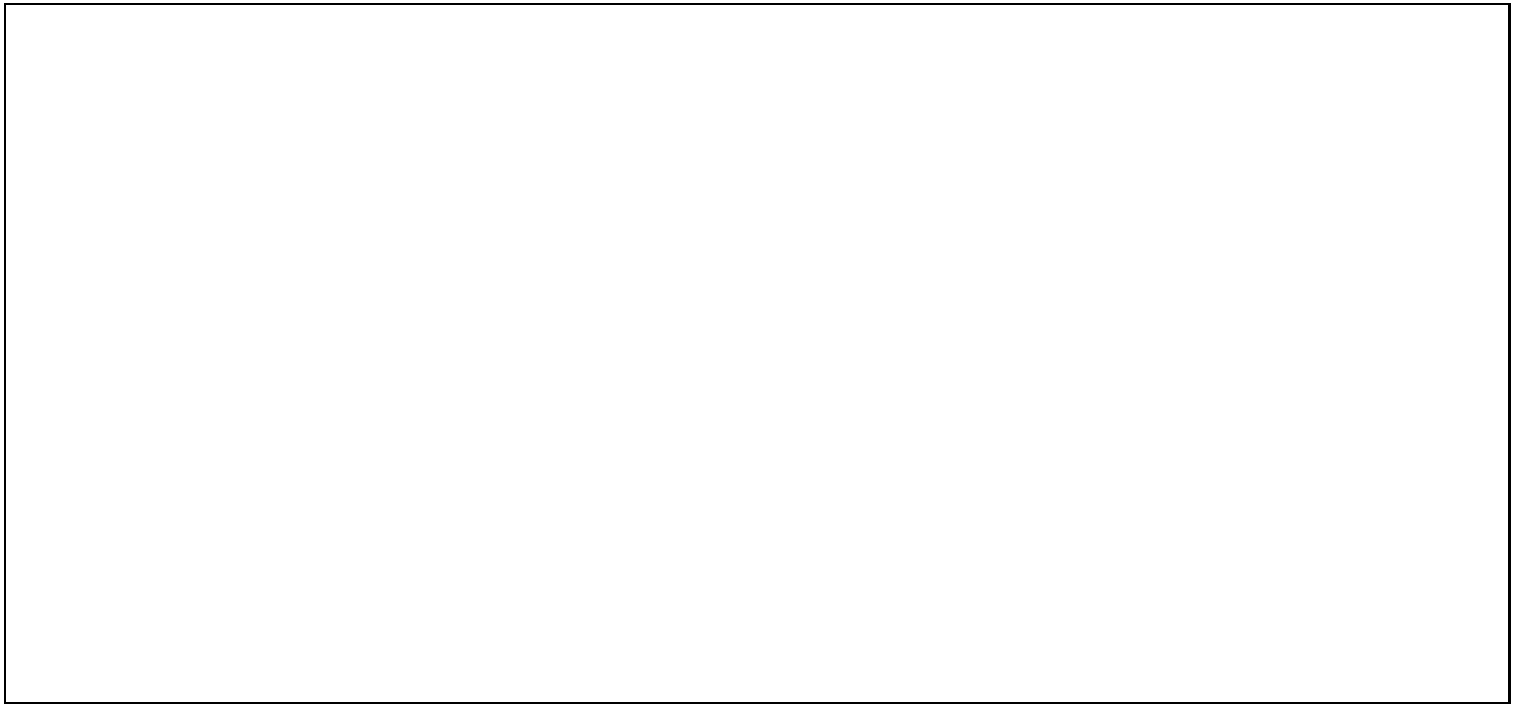
Particular Disease

Giving the result not only in person is healthy or not but also providing a particular disease which he/she can be infected to.

Space Complexity

To maintain the accuracy with a minimize version of dataset also.

CHAPTER 10: **REFERENCES**



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